Trade-Sage Innovation Report

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Project Overview

Trade-Sage is a production-quality equity trading signal prediction and portfolio risk scoring API. It uses an ensemble ML model (XGBoost + LightGBM + RandomForest VotingClassifier) to predict short-term bullish/bearish signals from OHLCV time-series data, and computes portfolio risk metrics including VaR, Expected Shortfall, Sharpe ratio, and maximum drawdown.

Tech Stack

- API: FastAPI + Pydantic v2, rate limiting, GZip, Correlation ID middleware
- ML: sklearn Pipeline, XGBoost, LightGBM, RandomForest VotingClassifier
- Features: 22 engineered (RSI, MACD, Bollinger, ATR, lag/rolling returns, price ratios)
- Database: SQLAlchemy ORM, Alembic migrations, PostgreSQL (prod) / SQLite (dev)
- Monitoring: KS-test drift detection per feature, 500-sample reference window
- Cache: Thread-safe TTL/LRU (512 entries, 300s TTL)
- Infra: Docker, docker-compose, GitHub Actions CI matrix (Python 3.11 + 3.12)
- Optional: MLflow stub, boto3/S3 stub, FAISS similarity index, Airflow DAG

API Endpoints (6 total under /api/v1/)

POST /predict - Ensemble signal: BUY/SELL + confidence + SignalStrength
POST /portfolio/risk - VaR, CVaR, Sharpe, max drawdown, avg correlation
POST /drift - KS-test drift detection across all 22 features
POST /train - Champion/challenger model retraining trigger
GET /health - Liveness + readiness probe with component checks
GET /metrics - Counters: predictions, drift alerts, cache hit rate, uptime

Test Results

166 passing, 4 skipped (faiss not installed in CI), 0 failures
16 test modules, 100% ruff lint compliance

Additional Modules

- app/backtest.py - Long-only signal backtester (equity curve, Sharpe, win rate)
- app/feature_importance.py - Ensemble feature importance extractor
- app/preprocessing.py - OHLCV validation, NaN filling, outlier clipping
- app/config.py - Centralised AppConfig with env-var overrides
- app/health.py - HealthStatus dataclass for liveness/readiness probes
- app/signals.py - SignalStrength enum + TradingSignal dataclass
- app/metrics.py - Thread-safe ServiceMetrics counters
- app/cache.py - TTL/LRU OrderedDict cache
- app/faiss_index.py - FAISS IndexFlatL2 with brute-force fallback

Repository Details

Branch: innovation/trade-sage

PR: https://github.com/atharvadevne123/reflective-lantern/pull/106

Commits: 60 (feat, test, docs, chore types)

Tag: trade-sage-v1.0.0